n.L → supervised n.L → Target x y

K N N

P = 4
having four parameters

Algorithm
Input → Data → Train → Test
x y

K = 5
① for each data point in Train
→ x_{test} (feed test pts) → Compute distance ($x^{(i)}$, x_{test})
dist → euclidean
i.e. list Cappend

② sort your distances (asc)
③ pick the k points based on the distances.
④ Pick majority

k=5 → 3 stand
2 -ive

KNN → supervised
→ regression
→ Classification (multi)

k=4 → 2 +ive
k=5 → 2 +ive, 3 -ive
n < labels
(k should not be multiple of j)

(parameters) / weights → Train → O(1)
(lazy learner) (non-parametric)

Da - advantage - prediction time
train (x) → d (features) VVV high

Pred pt
① compute dist → O(n d)
② sorting → O(n log n)
③ top k dist → O(k)
Total → O(n d + n log n + k)

for d prediction → O(n (d + log n))

logistic reg → $z = w_0 + w^T x$
O(d)

Advantage -
- simple ← intuitive
- Non linear - powerful

Assumption
- similar data points are together
(Homogeneous)

(+ve) x_q → +ve

Issue?

weighted k n n
$w^{(i)} = \frac{1}{d^{(i)}}$

-ive → 0.1 → 0.1 (d) $c_1 y_1 y^2$
+ive → 0.5 (d) $c_2 y_2 y^2$

$\frac{1}{0.1} = 10$
$\frac{1}{0.5} = 2$

Total weightage +ive
10+10
= 20

Total weightage -ive = 2
2+2+2 = 6

K means → dimensionality → reduction

euclid. dist. x_i, x_c → $\sqrt{\sum_{j=1}^{d} (x_{ij} - x_{cj})^2}$

↳ this is very slow if y and d is high.

Man hattan dist

$d = \left[\sum_{j=1}^{d} |x_{ij} - x_{cj}|\right]^1$

euc. dist

$|x_i - x_c| + |y_i - y_c|$
cosine

Cosine sim boiling

distance function → higher parameter

$d(x_i, x_c; p) = \left[\sum_{j=1}^{d} |x_{ij} - x_{cj}|^p\right]^{1/p}$

P=1 → man hattan (L) dist
P=2 → euclidean L2 dist

Bias - variance tradeoff.
overfit → fitting noise → high variance
underfitting → not even understanding the pattern → high bias

K

k=1
↓ -ive
↳ fitting noise
↳ high variance

k=7
majority → +ive

---

total no. of point → 100 → 60 +ive
↳ 40 -ive

x_q k=95 → always feedout majority (other)
↳ over fitting atmost $\frac{1}{2}$ of
↳ underfitting.

k ↓↓ → over fitting
k ↑↑ → under fitting.

Outliers

k=1 ↓↓
impact of outlier is high
when k=5

Interpretability

log reg → weights w_j

Pima Diabetes / healthy

x_q

k=5

using NEIGHBORS for interpretability

Pad=1

---

nearest neighbours
↳ features → numerical

categorical columns → 0 + 1 → ↑ dim increase
→ Target / non / learn
label

male, female P(y=1 | Gender, Exercise, Alcohol)
↳ 0.82 → male
0.82

Cosine similarity (x_q, x_c)

length $x_q - x_c$ length
$|x_q| \, |x_c|$

Probability of every label

k=5
4
1
dim

$P(y=1 | x_q) = \frac{4}{4+1}$
$P(y=0 | x_q) = \frac{1}{4+1}$

---

weighted KNN

+ive
w_1
dim

-ive
$w_2 \; w_3 \; w_4 \; w_5$
dim

$P(y=1 | x_q) \Rightarrow \frac{\sum_{+ive} w^{(i)}}{\sum_{all} w^{(i)}}$ → sum of weights of +ive neighbor / all weights

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Impute → missing data

SBI → bank bal → 0
↳ age → mean
↳ median
↳ intelligent imputation
↳ KNN imputation

N.N

x_q (impute) → J.M

k=5
mean
↳ x_q, x_c (feature a missing) → $\sum_{j=1}^{d} (x_j^{(1)} - x_j^{(2)})^2$